Darrin Bright

Medical Image Analysis | **Computer Vision** | **Deep Learning**

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Research Interests

Medical Image Analysis | 3D Vision | Label-Efficient Learning

Education

Vellore Institute of Technology (VIT) Vellore, India

Integrated Master of Technology in Software Engineering, CGPA: 8.81/10 *Sep 2022 - May 2027*

Relevant Coursework: Artificial Intelligence, Machine Learning, Statistics for Engineers, Soft Computing, Data Mining Techniques, Applied Discrete Mathematical Structures, Applications of Differential and Difference Equations.

Research Experience

Nanyang Technological University Singapore

Research Intern | Advisor: Dr. Si Yong Yeo *Jul 2026 - Present*

- Developing semi-supervised coronary vessel segmentation in X-ray angiography, addressing the limited availability of annotations in interventional imaging.

Simon Fraser University Vancouver, Canada

Research Intern | Advisor: Dr. Ghassan Hamarneh *Dec 2025 - Present*

- Developed a segmentation-free deep learning framework for detecting ER-mitochondria membrane contact sites directly in 3D super-resolution (STED, SIM) microscopy.
- Addressed the absence of ground truth with a physics-informed synthetic data pipeline, organelle geometry generated at 4 nm EM resolution, contacts labelled with an established EM contact definition and realistic images rendered with the pySTED and mcSIM microscope simulators.
- Validated against cell-line-specific contact biology (HT-1080 vs COS-7), MCS-DETECT and EM-derived ground truth (CLEM), showing spatial agreement across all three.

Indian Institute of Technology Kharagpur Kharagpur, India

Research Intern | Advisor: Dr. Debanjan Das *May 2026 - Jul 2026*

- Developed GraphUNet, representing beat-level ECG/PPG morphology as graphs within a U-Net architecture for cuffless blood-pressure estimation, enabling explicit modeling of physiological relationships across heartbeats.
- Evaluated on VitalDB and PulseDB against published baselines, achieving 4.97 mmHg SBP MAE and 2.77 mmHg DBP MAE under subject-disjoint, calibrated evaluation, demonstrating competitive cuffless BP estimation performance.

Indian Institute of Technology Hyderabad Hyderabad, India

Research Intern | Advisor: Dr. C. Krishna Mohan *Jan 2026 - Jun 2026*

- Built a federated continual learning framework integrating a global synthetic-data anchor with V-Information-guided selective re-distillation to preserve prior-task knowledge without sharing patient data or maintaining a conventional replay buffer.
- Evaluated across five MedMNIST modalities, improving average final accuracy by 32.12 percentage points and reducing average forgetting by 20.26 percentage points over FedCIL.

Vellore Institute of Technology Vellore, India

Undergraduate Researcher *Jul 2025 - Jun 2026*

Histology-Aware Framework for Oral Cancer Detection *Advisor: Dr. Bharanidharan N.*

- Developed SPHINX-Net, a lightweight hybrid CNN-Transformer framework for patch-level oral squamous cell carcinoma detection, combining histology-aware multi-view contrastive pretraining, spectral channel recalibration, and mini-batch prototype-graph refinement.
- Achieved 96.30% patch-level accuracy at 1,034 FPS with a lightweight 1.75M-parameter model, with external validation accuracies of 95.4% and 98.2% on lung, colon and colorectal tissue datasets, respectively.

PortionNet: Distilling 3D Geometric Knowledge for Nutrition Estimation *Advisor: Dr. Kanchan Keisham*

- Developed PortionNet, a cross-modal knowledge distillation framework that transfers 3D geometric knowledge from point clouds to an RGB-only model through a lightweight RGB-to-Geometry Adapter, eliminating depth-sensor requirements at inference.
- Achieved SOTA on MetaFood3D, with 17.43% volume MAPE and 15.36% energy MAPE, reducing errors by 57.9% and 77.4% relative to the previous best; further achieved 12.17% energy MAPE on the SimpleFood45 dataset.

Indian Institute of Technology Jodhpur Jodhpur, India

Research Intern | Advisor: Dr. Deepak Mishra *May 2025 - Jul 2025*

- Developed a hierarchical vector quantization framework for skin lesion segmentation, pretraining on unlabeled ISIC 2018 dermoscopy images and using the learned discrete codes to regularize PH2 fine-tuning: 79.76% Dice, a consistent +1.38% over the VQ-free baseline across repeated runs.
- Ran a 13-configuration ablation of vector quantization inside Swin Transformer encoders on Pascal VOC 2012, achieving 74.61% Dice, +7.34% over the CNN baseline and +3.91% over the unquantized Swin.

Publications

PortionNet: Distilling 3D Geometric Knowledge for Food Nutrition Estimation
11th Annual Conference on Vision and Intelligent Systems (CVIS), 2025 [Paper] [Code]

A Lightweight Histology-Aware Learning Framework with Spectral Channel Recalibration and Prototype-Graph Refinement for Oral Squamous Cell Carcinoma Detection
Results in Engineering, Under Review

Technical Skills

Programming Tools	Python | SQL | Git | LaTeX
ML & DL Tools	PyTorch | TensorFlow | OpenCV | Matplotlib | NumPy | Keras | MONAI
Medical Imaging	X-ray | CT | Histopathology | Fluorescence Microscopy | Dermoscopy

Work Experience

Astute Vellore, India

Project Manager and Technical Lead, Intern *Oct 2024 - Apr 2025*

- Architected SEO AgentFlow, a multi-agent AI framework with six specialized agents for keyword extraction, on/off-page optimization, technical SEO and website analysis.
- Led development of AI-driven automation workflows integrating RAG-based content generation, web scraping, and automated content and advertisement generation.

GenAI/ML Engineer, Intern *Jul 2024 - Oct 2024*

- Built a data-driven keyword research pipeline integrating SERP API, Google Trends, and DataForSEO with multi-dimensional ranking for competitive analysis.
- Developed automated advertisement generation using Stable Diffusion for poster design and Gemini for tagline generation and placement.

Leadership and Service

Chairperson, IEEE Robotics and Automation Society, VIT *Dec 2024 - Dec 2025*
Volunteer, National Service Scheme *Dec 2023 - Dec 2024*
Volunteer, Becoming I Foundation *Dec 2024 - Dec 2025*

Honors and Awards

Winner, Project 2039 Hackathon, Vellore Institute of Technology, 2024
Winner, Alphaforge Ideathon, Vellore Institute of Technology, 2024
Runner-up, Startup Demo Day, Vellore Institute of Technology, 2025
Runner-up, Biomimicry Innovation Challenge, Vellore Institute of Technology, 2024
Second Runner-up, HackWar Hackathon, Vellore Institute of Technology, 2025